

Problem 1: Two players alternate taking 1, 3, or 4 counters. Taking 2 is not allowed. Never take more than remain. Taking the last counter wins. Play from 8, 10, and 14. Record your moves. Swap who starts for a second attempt at a pile that surprised you.

Starting pile	First move	Next moves, in order	Winner: first or second
8			
10			
14			

Problem 2: W means the next player can force a win; L means every legal move gives the opponent a winning strategy. Zero is L. Build small piles and work upward: mark W if a legal move reaches L, and L if every legal move reaches W. Record the next piles as well as the labels.

Pile	Legal next piles	W/L
0	none	L
1		
2		
3		
4		
5		
6		
7		

Problem 3: Continue using moves 1, 3, and 4. Fill the chart in order. In each cell write L, or write W and a winning move: W1 means W with winning move 1. Check all legal next piles before marking L. Compare the rows after filling at least two of them.

Pile	0	1	2	3	4	5	6
Label/move							
Pile	7	8	9	10	11	12	13
Label/move							
Pile	14	15	16	17	18	19	20
Label/move							
Pile	21	22	23	24	25	26	27
Label/move							

Problem 4: Compare these four piles. Record every legal next pile and its label. Circle the starting piles that have no move to L.

Pile	Take 1: next pile/label	Take 3: next pile/label	Take 4: next pile/label
14			
16			
15			
17			

The repeating pattern I see:

Problem 5: Arrange 14, 16, 15, and 17 counters in whole groups of 7 with a smaller leftover group. Record the leftovers. Match each pile to its label in Problem 4.

Pile	Whole groups of 7	Leftover counters	W/L
14			
16			
15			
17			

Problem 6: Use your chart to choose a winning move for each leftover count marked W. Record the leftover count after the move. For counts marked L, list every leftover count an allowed move can reach; use piles 7 and 9 so that all three moves are possible.

Leftover count	W/L	Move to L, or all moves from L	New leftover count(s)
0			
1			
2			
3			
4			
5			
6			

Problem 7: Build 24 counters. Choose a first move using your table, then play. After each of your turns record the leftover count when grouping the remaining pile in sevens.

First move and later leftover counts:

Problem 8: Change the allowed moves to 1, 3, and 5. Play from 6 and from 7, then swap who starts. Build smaller piles to check a proposed strategy. Record which piles you want to hand the other player.

My games and the piles I want to leave:

Problem 9: Return to moves 1, 3, and 4. Copy the labels for piles 0–3 into the first row below and 7–10 into the second. These are four-label windows. To find the next label, inspect the first, second, and fourth cells: mark the next W if any is L, and L if all three are W. Record the next label, then slide the window one cell right.

Window starts at	Four labels	Next label	Window after sliding
0			
7			
1			
8			

Problem 10: Find another matching pair of four-label windows in your chart. Copy both windows, extend them by two labels, and compare the results.

My matching windows and their next two labels: